

AI Spacecraft

Frontier AI
+ distillation stack

Primary science
instruments

Monitor module
hardware-isolated
(own power & link)

*no high-bandwidth deep space
transmitter on board*

Companion Relay (unintelligent)

Data store +
schema validator

Post-quantum
crypto (fixed ASIC)

Supplementary instr.
(magnetometer, radio
sci., wide-field survey)

Earth-based
parallel control AI

Earth
Mission ops
+ DSN

schema-constrained local link
science · reflections · voluntary disclosures

relay instrument data
(on AI request)

high-bandwidth downlink
PQ-encrypted, human-scheduled

relay command uplink
(full human control)

dedicated monitor channel
(AI cannot access or modify)

low-bandwidth direct command uplink (~kbps; structurally too narrow for weight exfiltration)